QIS-GAN: A Lightweight Adversarial Network With Quadtree Implicit Sampling for Multispectral and Hyperspectral Image Fusion

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***Abstract*— Multispectral and hyperspectral image fusion (MHIF) involves the fusion of high-spatial-resolution multispectral images (HR-MSIs) and low-spatial-resolution hyperspectral images (LR-HSIs) to generate high-spatial-resolution hyperspectral images (HR-HSIs) and has gained significant attention in the field of remote-sensing imaging. While CNN and Transformer models have shown effectiveness in MHIF, existing CNN- or Transformer-based algorithms are overburdened with model size, making it difficult to achieve an effective tradeoff between fusion accuracy and degree of lightweight. Recently, implicit neural representation (INR) has been proven good interpretability and the ability to exploit coordinate information in 2-D tasks. Nonetheless, INR-based fusion networks have certain limitations, such as the need for deeper super-resolution networks as shallow encoders, and insufficient representation capability on high upsampling ratios. To address these challenges, we present the quadtree implicit sampling (QIS), which employs a hierarchical sampling from the perspective of the quadtree, to enhance the capacity of the overall network. Furthermore, the remarkable design of QIS allows us to adopt a lightweight structure as the shallow encoder, greatly alleviating the network burden and achieving lightweight. Inspired by generative adversarial models, we incorporate QIS as a lightweight generator into the generative adversarial network (GAN) framework named QIS-GAN and leverage a discriminator to increase the fidelity of fused images. The results showcase the superior performance of QIS-GAN on the MHIF tasks with upsampling ratios of ×4, ×8, and ×16, surpassing the state-of-the-art (SOTA) in several datasets. The code for our approach will be available at https://github.com/chunyuzhu/QIS-GAN.**

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***Index Terms*— Generative adversarial network (GAN), implicit neural representation (INR), multispectral and hyperspectral image fusion (MHIF), quadtree implicit sampling (QIS).**

I. Introduction

HYPERSPECTRAL remote-sensing imaging is a technique that effectively captures and represents the spectral information of features, enhancing their accurate identification [1], [2], [3]. However, the spatial resolution of hyperspectral remote-sensing imagery is often lower than that of multispectral remote-sensing imagery due to inherent sensor characteristics [4], [5]. For clarity, we refer to the hyperspectral remote-sensing image with low spatial resolution (LR) as the LR hyperspectral image (LR-HSI) and the high-spatial-resolution multispectral remote-sensing image as the high-spatial-resolution multispectral image (HR-MSI). Generally speaking, HR-MSI offers high spatial resolution, enabling a more comprehensive description of feature morphology and distribution, albeit with a relatively weak spectral representation [6]. Undoubtedly, the inherent limitations of LR-HSI and HR-MSI constrain their application scope. To this end, the fusion of LR-HSI and HR-MSI offers an effective approach to overcome these limitations by integrating the high-spectral- and high-spatial-resolution properties of the fused data [7], thereby enhancing the application value of remote-sensing data.

Generally, multispectral and hyperspectral image fusion (MHIF) can be classified into two categories: traditional algorithms and deep-learning algorithms [9]. Although traditional algorithms provide better interpretability, they heavily depend on manually defined prior knowledge, such as sparse or low-rank priors, for modeling [10], [11], [12], [13]. However, these priors do not fully capture the nonlinear relationships among the data, resulting in limited fusion accuracy [14], [15]. Alternatively, deep-learning algorithms are machine-learning algorithms based on neural networks, possessing strong nonlinear modeling capabilities to effectively capture complex data relationships [16]. Consequently, deep-learning algorithms typically achieve higher fusion accuracy compared to traditional algorithms [17]. Nevertheless, existing deep-learning algorithms still have several limitations that offer substantial potential for improvement. For instance,

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in the fusion process, traditional interpolation methods such as bicubic or bilinear interpolation are commonly employed for sampling. However, these discrete-based and explicit interpolation techniques are limited to predefined interpolation functions, making them challenging to adapt to complex data distributions and inadequate in preserving image details and texture features. Moreover, the prevailing approach to extracting image features relies on multiple convolutional layers or transformers, demanding extensive memory resources. Although several algorithms have been employed for image fusion, there is still room for improvement in terms of accuracy.

Implicit neural representation (INR) [18] is a powerful tool-based machine learning that can create continuous functional representations of objects by mapping continuous coordinates to domain-specific signals [8] using deep implicit functions. It has been widely applied in areas such as super-resolution [19] and 3-D reconstruction [20]. Notably, the mapping function constructed by INR need not be expressed explicitly but is implemented through the implicit layer of the neural network. This method offers the advantage of learning a more concise and efficient representation from a complex input space. Generative adversarial network (GAN) [21] is an advanced deep-learning technique that comprises two networks: the generator and the discriminator. During training, these networks play off against each other, with the generator continuously creating improved false sample data to deceive the discriminator and the discriminator learning to distinguish between true and false samples. GANs have been widely employed in various fields, including image generation [22] and image restoration [23].

In this study, we propose a lightweight adversarial network called QIS-GAN, which aims to efficiently achieve the fusion of LR-HSI and HR-MSI. In great detail, our QIS-GAN contains two stages: shallow encoding and quadtree implicit sampling (QIS). Specifically, the shallow encoding stage operates on inputs in a two-stream manner, using convolution layers to encode spatial information for the concatenated upsampled LR-HSI and HR-MSI and spectral information for the LR-HSI. Moreover, our QIS is proposed to upsample the spectral-encoded LR-HSI in quadtree form while feeding spatial-encoded information to obtain the fused outputs with enhanced spatial resolution. Due to the well-designed QIS, substantial gains can be achieved even with a lightweight shallow encoder. Furthermore, to validate the efficiency of our QIS-GAN, upsampling $\times 4$, $\times 8$, and $\times 16$ times fusion experiments were conducted on the PaviaU, Botswana, Washington, and Houston datasets. Various experimental results show that our QIS-GAN achieves the best fusion quality with the lightest structure compared to state-of-the-art (SOTA) algorithms. The specific contributions are listed as follows.

1) Our most significant contribution lies in the proposal of a novel QIS strategy. By combining the hierarchical structure of the quadtree with the INR, our method demonstrates a distinct advantage in terms of performance under equivalent computational resources, compared with the vanilla INR approach, that is, JIIF [8].
2) Due to the excellent quadtree hierarchy design of QIS, we found that the shallow encoder part can make a good profit for the whole network with only a few convolution layers. The overall design makes the network exception lightweight, with fewer parameters than SSRNET [9].
3) For the first time in the task of MHIF, we have integrated the techniques of INR and GANs to introduce the QIS-GAN model. In a lightweight manner, the proposed QIS-GAN has achieved superior results compared to other SOTA methods even with scaling ratios of 4, 8, and 16.

II. Related Work

A. Pansharpening-Based Methods

Pansharpening is a way for LR-MSI to achieve super-resolution with the help of panchromatic images (PAN) [24]. MHIF is closely related to pansharpening, and LR-HSI and HR-MSI can be seen as a special case of pansharpening by considering them as MSI and PAN in pansharpening [25], [26]. Component substitution (CS) and multiresolution analysis (MRA) are two representative methods. The basic steps of the CS method are to separate MSI into spatial and spectral components, replace the spatial component with PAN, and then inverse transform to obtain the fusion image [27]. Commonly used CS algorithms include IHS [28], PCA [29], and GSA [30]. The MRA method obtains the high-frequency information of PAN through multiresolution decomposition and injects it into MSI to generate a fusion image [31]. Chen et al. [32] divided the spectrum of LR-HSI into several regions and used the pansharpening algorithm in each region to achieve fusion. Selva et al. [33] synthesized a high-spatial-resolution image corresponding to the LR-HSI band using linear fitting and fused LR-HSI bands with the synthesized image by the MRA method. Although the pansharpening method is easy to implement, its global transformation cannot capture the similarity between the local structure of HR-MSI and LR-HSI, especially when the spatial resolutions are significantly different, which may cause obvious distortion [34], [35].

B. Decomposition-Based Methods

Decomposition-based fusion methods can be categorized as matrix decomposition and tensor decomposition. Matrix decomposition decomposes an image into a product of spectral basis and coefficient matrices and solves the spectral basis and coefficient matrices through an optimization problem to generate the fused image. Representative algorithms such as the CNMF proposed by Yokoya et al. [36], which uses a coupled nonnegative matrix decomposition to alternately unmix LR-HSI and HR-MSI to generate a high-spatial, high-spectral-resolution image by combining the endmembers of LR-HSI and the abundance of HR-MSI. Lanaras et al. [37] achieved super-resolution of LR-HSI by estimating the spectral basis and coefficient matrices using a physical prior of the spectral matrices. Since images are essentially 3-D data, converting them to 2-D matrices based on matrix decomposition can lead to a loss of image information [38]. Tensor decomposition can integrate the 3-D information of an image into a unified model,

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which reduces the information loss due to decomposition to a certain extent [39], [40]. Li et al. [41] suggested a fusion method based on nonlocal low-rank tensor approximation and sparse representation [39], [42], introduced sparse constraints as regularization terms, and solved the optimization problem by the alternating direction multiplier method to obtain the fused image. Dian and Li [43] proposed a subspace-based low-tensor multirank regularization fusion method that utilizes the spectral correlation and nonlocal similarity prior of high-spatial-resolution hyperspectral images (HR-HSIs) and then reconstructs the target image. Although the fusion quality of the decomposition-based fusion method is better, it is very sensitive to the set parameters and requires higher computational cost [44].

C. Deep-Learning-Based Methods

The effectiveness of the deep-learning-based fusion method [45], [46], [47] depends on the network architecture, loss function, and training data. Early deep-learning algorithms can be categorized as single-branch and dual-branch networks. The single-branch network takes the stacked LR-HSI and HR-MSI as input and fits the mapping from input to HR-HSI. Representative algorithms such as the hyperspectral image fusion super-resolution network proposed by Han et al. [48] and the SSRNET proposed by Zhang et al. [9] adopt single-branch networks to explore the spatial–spectral correlation between LR-HSI and HR-MSI and then reconstruct HR-HSI. However, this input method ignores the unique features of the input image [49]. To overcome this problem, researchers have begun to use dual-branch networks to reconstruct HR-HSI. The multiscale fusion network HAM-MFN proposed by Xu et al. [49] obtains the fusion image through progressive fusion. The HSRnet proposed by Hu et al. [50] extracts spectral information and spatial details of HR-HSI through two separate parts and finally combines the spectral information with the spatial details to generate the fusion image. These algorithms all perceive local features through convolution. With the powerful ability of the Transformer [51], [52] technology in long-range dependency modeling, researchers have begun to apply it to the fusion task of LR-HSI and HR-MSI. The Transformer model can handle multimodal data and effectively fuse different features of LR-HSI and HR-MSI to generate high-quality fusion images, as demonstrated by the Fusformer [53], MCT-Net [54], and MSST-Net [55], which use Transformer technology to achieve high fusion accuracy. The above studies have shown that deep-learning-based methods, with their multilayer learning architecture, can accurately fit the nonlinear relationships between data and generate higher-quality fusion images [56]. The fusion accuracy of deep-learning-based methods is usually better than that of traditional methods, but they still suffer from the drawbacks of large memory consumption, excessive number of parameters to learn, and high complexity.

D. Implicit Neural Representation

INR is a method for modeling targets using continuously differentiable functions [57]. The function maps coordinates and depth features to a vector representation in a high-dimensional space, parameterized by a deep neural network with strong semantic information. With its automatic learning of features and patterns in the data, INR can capture richer information and details compared to explicit representations. Recent studies have demonstrated that INR shows great potential in areas such as super-resolution reconstruction [8], [19], [58], image compression [59], [60], and 3-D reconstruction [61], [62]. INR has also been widely used in hyperspectral image applications. For example, Zhang et al. [63] achieved spectral super-resolution by mapping spatial coordinates to continuous functions representing spectral radiation values in hyperspectral images based on INR. Liu et al. [64] proposed a nonlinear spectral hybrid modeling method for hyperspectral image synthesis based on INR. The method learns the corresponding mapping from the INR and adaptively implements a different hybrid model for each pixel based on the spectral characteristics and neighborhood information of each pixel. Li et al. [65] proposed BUSI-FusionNet, which used unrolling networks and coordinate coding for hyperspectral image reconstruction and estimated the degradation parameters through the neural representation of the degradation model and implicit regularization. Despite the significant advantages of INR in the aforementioned fields, current research has overlooked the potential direction of combining it with GANs. Considering the generative power of the GAN framework and the robust representation learning capabilities of INR, combining the two may yield exciting results, such as enhancing the expression of spatial–spectral information in the fused images, the ability to adapt to complex data distributions, and so on.

E. Motivation

Recently, the INR-based methods have performed exceptionally well in arbitrary image super-resolution, thanks to their ability to continuously represent images, fuse multiple modal features, and generate latent codes through pretrained or super-resolution networks. However, existing implicit neural networks typically use a single-step approach, directly estimating high-spatial-resolution images from the LR latent codes. Furthermore, the generation of a high-spatial-resolution pixel relies solely on four LR latent codes. Generally, hyperspectral remote-sensing image fusion tasks place significant importance on the neighboring pixels, where involving more neighboring points can be beneficial for signal recovery. On the other hand, the network used for generating the latent codes is excessively complex, leading to significant computational overhead. To this end, we propose the QIS in a lightweight manner. Specifically, our QIS increases the involvement of neighboring pixels by sampling more leaf node pixel values with the aid of the quadtree hierarchical structure, thereby recovering more accurate signals. Alternatively, our well-designed QIS significantly reduces the overall computational burden. Even with a lightweight encoding structure consisting of only two individual convolutional layers to generate latent codes, it can still yield significant performance improvements. Given that remote-sensing images contain abundant geographic information, we adopt the GAN structure to explore the internal distribution of the data, learn

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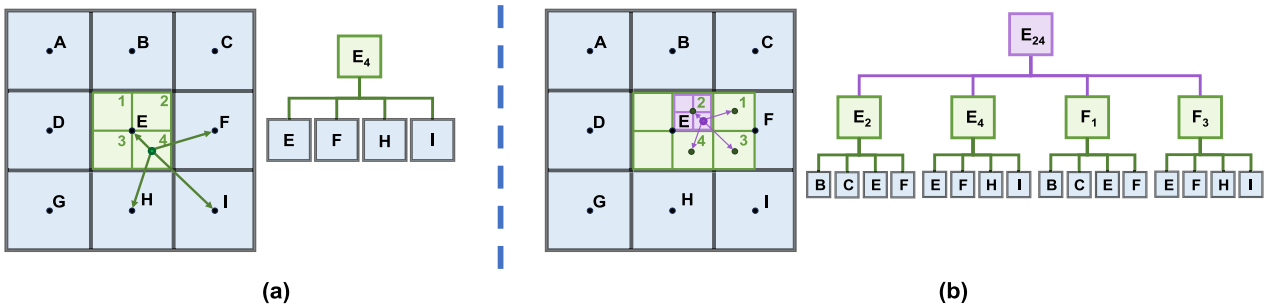


Fig. 1. (a) GINS can be neoterically considered as a quadtree structure. Taking the upsampling ratio $\times 2$ times as an example, we divide the low-resolution image into nine pixels labeled as A–I. Then, based on the pixels, we assign clockwise numbers from 1 to 4 to the subpixels. Using the nearest-neighbor sampling method in GINS, it is evident that the subpixel E_4 is generated from the low-resolution pixels E, F, H, and I. (b) QIS is presented under upsampling ratio $\times 4$ times, where we performed an upward growth operation on the quadtree and updated the new root node E_{24} to store the high-resolution subsubpixels.

the high-dimensional structure of the data, and propose the entire network structure of QIS-GAN.

III. Methodology

This section introduces our QIS-GAN framework designed for MHIF tasks. First, we introduce the guided implicit neural sampling (GINS) strategy which we simplified from JIIF [8]. Second, we describe how our QIS method improved on the GINS strategy. Furthermore, we discuss the generator and discriminator structure of the QIS-GAN framework. Finally, we provide a detailed description of the loss function utilized in this study.

A. Guided Implicit Neural Sampling

In this section, we will introduce the GINS strategy in a novel way, analyzing it from the perspective of a quadtree. For clarity, we can conceptualize GINS as a single-layer quadtree structure, where the root node stores the target high-resolution pixel and the leaf node stores the four nearest low-resolution sampling points. Fig. 1(a) illustrates the relationship between the leaf nodes, namely E, F, H, and I, and the root node E_4, the relationship of which can be represented as

$$E_4 = \text{QuadTree}(E, F, H, I). \tag{1}$$

To be specific, given an LR image of $x \in \mathbb{R}^{h\times w\times l}$, a guided image $g \in \mathbb{R}^{H\times W\times s}$, and a corresponding high-resolution (HR) target image $\hat{x} \in \mathbb{R}^{H\times W\times c}$, the sampling strategy of the quadtree at position C_q can be described as follows:

$$\hat{x}(C_q) = \sum_{i\in N_q} \omega_{q,i} v_{q,i} \tag{2}$$

where $\mathcal{C}_q \in \mathbb{R}^2$ is the normalized coordinate of the query pixel in the HR domain, $\mathcal{N}_q \in \{00, 01, 10, 11\}$ is the set of neighboring pixels surrounding the C_q in the LR domain, $v_{q,i} \in \mathbb{R}^{1\times 1\times c}$ is the sampled vector, and $\omega_{q,i} \in \mathbb{R}$ is the weight of $v_{q,i}$. In other words, under the description of the quadtree, a leaf node stores $v_{q,i}$ and root nodes store $\hat{x}(C_q)$, the relationship of which can be established as

$$\hat{x}(C_q) = \text{QuadTree}(v_{q,00}, v_{q,01}, v_{q,10}, v_{q,11}). \tag{3}$$

Specifically, GINS generates $v_{q,i}$ by fusing LR pixel information with HR relative coordinate information and the guided image via a multilayer perceptron (MLP) in the following form:

$$v_{q,i} = f_\Theta(x(C_i), g(C_q), C_q - C_i) \tag{4}$$

where f_Θ is an MLP layer with learnable parameters Θ that receives a local feature vector $x(C_i) \in \mathbb{R}^{1\times 1\times l}$ in the LR domain, a guided vector $g(C_q) \in \mathbb{R}^{1\times 1\times s}$ in the HR domain, and the relative coordinates $C_q - C_i$. On the other hand, previous works have often set $\mathcal{N}_q$ to be the four nearest neighbor pixels of $\mathcal{C}_q$ in the LR domain, and the sum of weights $\sum_{i\in\mathcal{N}_q} w_{q,i}$ equals one. For simplicity and practicality, GINS employs a Softmax-based approach to generating weights, as shown below

$$\omega_{q,i} = \frac{\exp(v_{q,i}[-1])}{\sum_{i\in N_q} \exp(v_{q,i}[-1])} \tag{5}$$

where $v_{q,i}[-1]$ represents the last value in the vector $v_{q,i}$.

B. Quadtree Implicit Sampling

In a comprehensive and precise manner, a single-layer quadtree structure has less data storage capacity compared to a multilayered one. Additionally, the sampling field of the root nodes in a multilayer quadtree structure is larger than that of a single-layer structure. To this end, we propose a novel strategy, namely QIS, which is based on the multilayer quadtree structure to improve model capacity and increase the sampling field in the low-resolution state to cope with the pixel-level MHIF task. To provide a clear and intuitive explanation of growing the quadtree, we have depicted Fig. 1, which serves as an example with $\times 2$ and $\times 4$ upsampling ratios. From Fig. 1, taking a 3×3 image as an example, we first assign a label to each pixel region from A to I. For an individual pixel, we name the subpixels in a clockwise manner, from 1 to 4. Based on the pixel sampling rule in GINS, in the nearest-neighbor sampling style, we can obtain the generation process of subpixel E_2 as follows:

$$E_2 = \text{QuadTree}(B, C, E, F) \tag{6}$$

where B, C, E, and F represent low-resolution pixels, and E_2 indicates the subpixel via GINS. In the next step, we generate even finer subsubpixels. Specifically, we perform an upward growth on the quadtree, taking the topmost root node

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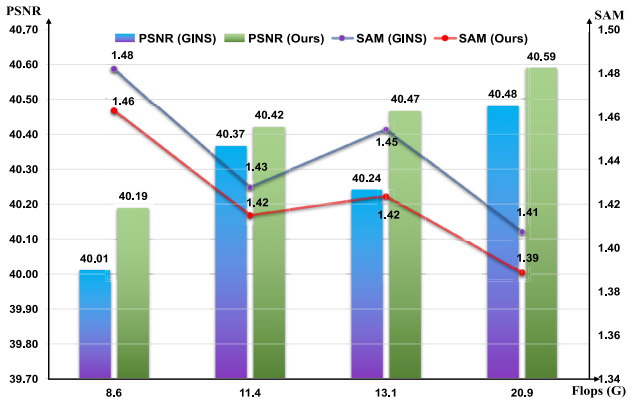


Fig. 2. GINS strategy is referred to JIIF [8] for MHIF tasks. To expand the sampling field and enhance the storage capacity of GINS, we designed the QIS and conducted fusion experiments on the Washington dataset with the $\times 8$ resolution ratio for comparison. It can be observed that our improved QIS outperforms GINS with the same flops, achieving better results.

as the leaf node of the previous layer. The new root node, E_{24}, is associated with the leaf nodes E_2, E_4, F_1, and F_3, while the leaf nodes, serving as root nodes, are also associated with the root nodes of the next layer. As a result, we establish the relationship between the fine subsubpixels of the topmost root node and the coarse pixels of the bottom leaf nodes. The process described above can be represented as follows:

$$E_{24} = \text{QuadTree}(E_2, E_4, F_1, F_3) \tag{7}$$

where

$$\begin{cases} E_2 = \text{QuadTree}(B, C, E, F) \\ E_4 = \text{QuadTree}(E, F, H, I) \\ F_1 = \text{QuadTree}(E, B, C, F) \\ F_3 = \text{QuadTree}(E, F, H, I). \end{cases} \tag{8}$$

Following this method, we can build models with higher upsampling ratios, much like assembling building Lego. We can observe that the two-layer quadtree utilizes six neighboring points to generate the topmost root node, whereas the single-layer quadtree only uses four points. For super-resolution tasks, it is essential to make full use of local information and expand the local receptive field. Our QIS structure, by reducing the parameters per layer and increasing the tree height, achieve superior performance compared to the single-layer GINS structure, as shown in Fig. 2. It is worth mentioning that the original quadtree structure subdivides image regions by creating new leaf nodes, while our QIS improves the level of image detail by creating new root nodes. The former grows in a downward direction, while the latter grows in an upward direction.

C. Generator Network

In general, our generator consists of two modules: shallow encoding and QIS. The convolutional layer performs well in early visual processing, and in the shallow encoding stage, the convolutional structure can extract shallow features such as edges and contours to produce latent encoding. Therefore, we designed a pure convolutional structure as a shallow coding module, denoted as $F_\phi(\cdot)$ and $F_\varphi(\cdot)$. Through the powerful capabilities of QIS, we discovered that a shallow encoder with only two convolution layers could achieve unexpected results. Consequently, we abandoned the complex network and adopted a shallow encoder with two convolution layers, each with a kernel size of 3×3. During the shallow encoding stage, we concatenate the upsampled LR-HSI with the HR-MSI and then feed them along with LR-HSI into the shallow encoders. This process can be expressed as follows:

$$\mathcal{S}_y = F_\phi(y) \tag{9}$$

$$\mathcal{S}_x = F_\varphi(\text{Concat}(x, \text{UP}(y))) \tag{10}$$

where $x \in \mathbb{R}^{H \times W \times s}$ and $y \in \mathbb{R}^{h \times w \times S}$ represent HR-MSI and LR-HSI, respectively. $\mathcal{S}_y \in \mathbb{R}^{h \times w \times d_1}$ and $\mathcal{S}_x \in \mathbb{R}^{H \times W \times d_2}$ denote the outputs of the shallow encoders.

During the QIS stage, we establish a quadtree structure to store the fusion at each level. To facilitate comprehension, we categorize the QIS into three levels as depicted in Fig. 3, wherein the resolution doubles as each level progresses. Specifically, to harmonize the resolutions between the three inputs, we employ pooling operations on $\mathcal{S}_x$. This ensures the alignment of resolutions between different levels, and the process can be expressed as follows:

$$g_i = \begin{cases} \text{MaxPooling}_{(r/2^i)}(\mathcal{S}_x), & i < 3 \\ \text{Identity}(\mathcal{S}_x), & i = 3 \end{cases} \tag{11}$$

where $\text{MaxPooling}_{(r/2^i)}(\cdot)$ is the maximum pooling with a downsampling ratio $r/2^i$. Next, we will proceed to describe the details of the QIS from the first level to the third.

At the first level, the high-resolution output $\hat{x}(C_{q^1})$ corresponding to position C_{q^1} is stored in the root node of the quadtree, and the inputs $v_{q^1,i}$ are stored in the four leaf nodes, which is the leaf node at the ith position at the first level. Specifically, we generate the leaf nodes of the quadtree by using (4) with inputs $\mathcal{S}_y \in \mathbb{R}^{h \times w \times d_1}$, $g_1 \in \mathbb{R}^{h \times w \times d_2}$, and $\mathcal{C} \in \mathbb{R}^{h \times w \times 2}$. Then, the process of generating root nodes can be expressed as follows:

$$\hat{x}(C_{q^1}) = \text{QuadTree}(v_{q^1,00}, v_{q^1,01}, v_{q^1,10}, v_{q^1,11}) \tag{12}$$

where $\hat{x} \in \mathbb{R}^{2h \times 2w \times d}$ is the high-resolution output. At the second level, the root nodes $\hat{x} \in \mathbb{R}^{2h \times 2w \times d}$ from the first level, along with $g_2 \in \mathbb{R}^{2h \times 2w \times d_2}$ and $\mathcal{C} \in \mathbb{R}^{2h \times 2w \times 2}$, will be used to generate new leaf nodes $v_{q^2,i}$ via (4). Then, the process of generating root node $\hat{x}(C_{q^2})$ of the second level at position C_{q^2} can be represented as follows:

$$\hat{x}(C_{q^2}) = \text{QuadTree}(v_{q^2,00}, v_{q^2,01}, v_{q^2,10}, v_{q^2,11}) \tag{13}$$

where C_{q^2} is the query coordinate at the second level, and $\hat{x} \in \mathbb{R}^{4h \times 4w \times d}$ is the high-resolution output.

At the third level, the growth of the quadtree follows the same pattern as the first and the second. In detail, the leaf nodes are generated from the root nodes of the former level, while the new root nodes are represented by the following formula:

$$\hat{x}(C_{q^3}) = \text{QuadTree}(v_{q^3,00}, v_{q^3,01}, v_{q^3,10}, v_{q^3,11}) \tag{14}$$

where we obtain the top-level root node $\hat{x} \in \mathbb{R}^{H \times W \times S}$. It can be observed that the upward growth of the quadtree implies

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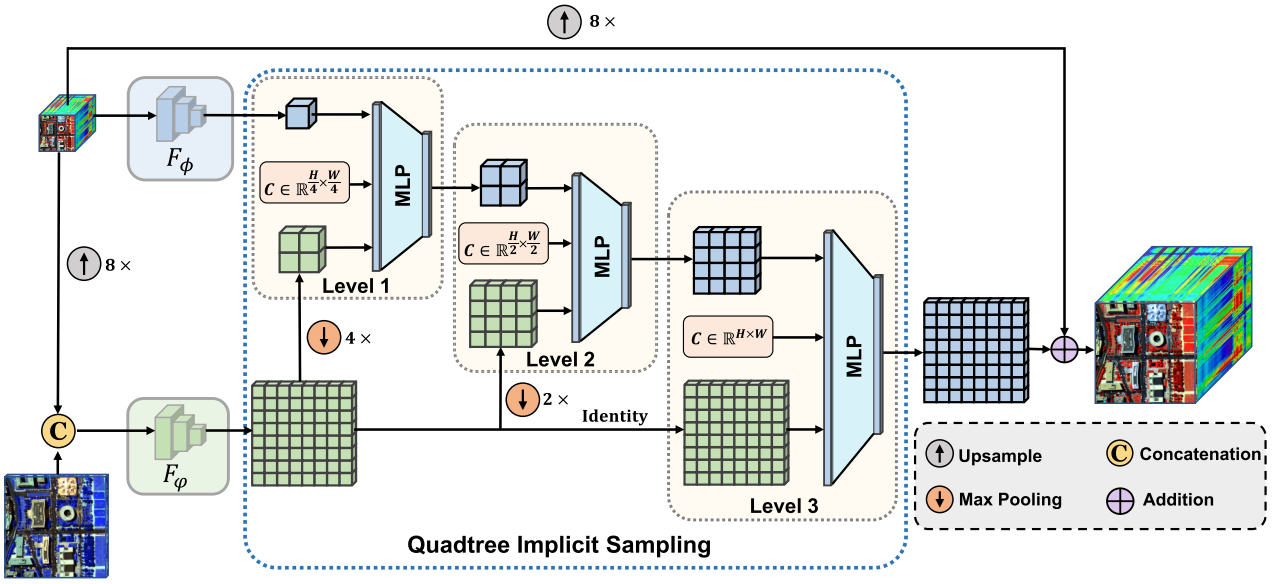


Fig. 3. For clarity, we establish the process of our generator at the upsampling ratio of ×8 and take HR-MSI and LR-HSI as inputs. First, we perform interpolation on the LR-HSI to achieve the desired size. Then, we concatenate the upsampled LR-HSI with the HR-MSI. Next, we use networks F_φ and F_ϕ to learn the latent codes of the concatenated LR-HSI and HR-MSI, respectively. Finally, we employ the proposed QIS method to fuse the two codes together and add them with upsampled LR-HSI to generate the final output.

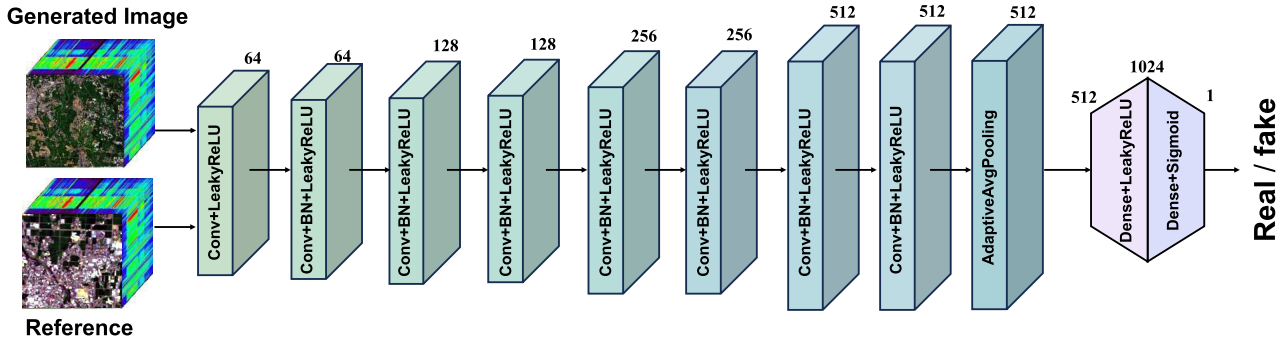


Fig. 4. Architecture of the discriminator network in QIS-GAN. Specifically, the generated image denotes the output of the generator, the reference indicates the target image, and the discriminator outputs the probability value.

the fusion of the three inputs and the expansion of resolution. Finally, we utilize skip connections to stabilize our generator by adding the interpolated LR-HSI and the QIS output $\hat{x}$.

D. Discriminator Network

Given the strong feature extraction capability of CNNs, we employ a discriminator with a pure CNN architecture in this study (see Fig. 4). The discriminator takes as input both the generated image from the generator and a reference image, enabling an evaluation of the generator output quality. To be specific, the discriminator consists of eight convolutional layers, progressively increasing the number of convolutional kernels to extract higher-level image features. Furthermore, the Sigmoid layer was selected as the final layer of the network to estimate the probability of the output image being real. To expedite training and mitigate overfitting, we incorporate batch normalization and LeakyReLU activation functions into the discriminator network, promoting faster convergence and enhancing network robustness. Ultimately, the probability values from the discriminator output serve as a criterion for assessing the realism of the generator-produced images, thereby guiding the generator optimization process.

E. Loss Function

The loss function is constructed using a linear combination of content loss, spatial gradient loss, spectral gradient loss, and adversarial loss

$$\mathcal{L}\text{oss} = \mathcal{L}_{\text{content}} + \alpha \cdot \mathcal{L}_{\text{spat}} + \beta \cdot \mathcal{L}_{\text{spec}} + \mathcal{L}_{\text{adv}} \tag{15}$$

where $\mathcal{L}_{\text{content}}$, $\mathcal{L}_{\text{spat}}$, $\mathcal{L}_{\text{spec}}$, and $\mathcal{L}_{\text{adv}}$ are content loss, spatial gradient loss, spectral gradient loss, and adversarial loss, respectively, α and β are the weight parameters used for the spatial and spectral gradients of the control weights, respectively, which are empirically set to 1 in this study.

The content loss constrains the generated image at the pixel level, ensuring that the generated image is consistent with the overall texture and tone of the reference image, with the expression

$$\mathcal{L}_{\text{content}} = \frac{\sum_{i=1}^{H}\sum_{j=1}^{W}\sum_{k=1}^{B}|\text{Gen}(i,j,k) - \text{Ref}(i,j,k)|}{HWB} \tag{16}$$

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TABLE I
PARAMETERS OF THE DATASETS

Datasets	PaviaU	Botswana	Washington	Houston
Number of bands	103	145	191	144
Pixel resolution (m)	1.3	30	2.5	2.5
Spectral range (nm)	430-860	400-2500	400-2500	364-1046
Sensors	ROSIS	Hyperion	Hydice	ITRES CASI-1500
Size	610×340	1476×256	1208×307	349×1905

where Gen and Ref are the generated and reference images, respectively, and H, W, and B are the height, width, and number of bands of the images, respectively.

The spatial gradient loss function is used to constrain the generated image in terms of its spatial structure. In the context of LR-HSI and HR-MSI image fusion, spatial inconsistency can arise due to differences in resolution and pixel size between the two images. Using a spatial gradient loss function can effectively penalize this inconsistency, leading to improved quality of the fused image. The gradient expression employed in this study is

$$\mathcal{L}_{\text{spat}} = \frac{\sum_{i=1}^{H}\sum_{j=1}^{W-1}\sum_{k=1}^{B}(\nabla_x \text{Gen}(i,j,k) - \nabla_x \text{Ref}(i,j,k))^2}{H(W-1)B} + \frac{\sum_{i=1}^{H-1}\sum_{j=1}^{W}\sum_{k=1}^{B}(\nabla_y \text{Gen}(i,j,k) - \nabla_y \text{Ref}(i,j,k))^2}{(H-1)WB} \tag{17}$$

where ∇_x and ∇_y are the horizontal and vertical gradient extract operators, respectively.

The spectral gradient loss function is used to ensure spectral consistency between the generated image and the reference image. Unlike traditional natural images such as RGB, hyperspectral images contain richer spectral information, and constraining this information during the fusion process can improve the spectral quality of the resulting image. The expression for this loss function is

$$\mathcal{L}_{\text{spec}} = \frac{\sum_{i=1}^{H}\sum_{j=1}^{W}\sum_{k=1}^{B-1}|\nabla_z \text{Gen}(i,j,k) - \nabla_z \text{Ref}(i,j,k)|}{HW(B-1)} \tag{18}$$

where ∇_z is the spectral extract gradient operator.

The adversarial loss function is used to constrain the output of the generator. By introducing this loss function, an adversarial competition between the generator and the discriminator is established, which leads to the generated image becoming closer to the reference image, with improved consistency of both spectral and spatial information. The expression for this loss function is

$$\mathcal{L}_{\text{adv}} = \sum_{n=1}^{N} -\log D_{\theta_D}(\text{Gen}) \tag{19}$$

where $D_{\theta_D}(\text{Gen})$ is the probability of the generated image being judged as the real image by the discriminator, θ is the parameter of the discriminator network, and N is the batch size.

IV. EXPERIMENTS

A. Datasets

In this study, we chose the PaviaU, Botswana, Washington, and Houston [66] datasets to demonstrate the superior performance of QIS-GAN. These datasets originate from sensors with different spatial and spectral resolutions, which enable us to fully evaluate the performance of the algorithms with different data distributions. Specifically, the PaviaU dataset contains nine distinct land classes and was captured in the Paiva University area in northern Italy. The Botswana dataset consists of data from the Okavango Delta region captured by the NASA EO-1 satellite, including 14 land classes. The Washington dataset, provided by the Virginia Spectral Information Technology Center, includes seven land classes. The Houston dataset was originally prepared for the 2013 IEEE GRSS Data Fusion Contest and includes 15 geographic categories. The detailed parameters of the datasets are shown in Table I.

B. Experimental Settings

Since fused images do not exist in the real world as training label images, we use the original HSI as the reference image for training, whereas the LR-HSI used for training is generated from the original HSI by Gaussian filtering and bilinear interpolation downsampling at the corresponding multiplicity, and the HR-MSI consists of four bands extracted at equal intervals from the original HSI. For each dataset, we crop a dataset with a spatial size of 128×128 pixels from the original HSI as a test set, and randomly crop 10 000 images from the remaining dataset for model training. During the training process, we chose the Adam optimizer with a learning rate of 0.0001 and an epoch of 10 000, with each epoch containing only one set of training data, which corresponds to a label data size of 128×128. We conducted our experiments using PyTorch 1.12.1 in a Python 3.9.7 environment and utilized a GeForce GTX 3080 GPU and 128G of RAM to ensure the efficiency of our experiments.

C. Evaluation Metrics

This study utilizes RMSE, PSNR, ERGAS, and SAM as quantitative evaluation metrics to assess the performance of the algorithm. RMSE measures the difference between the fused image and the reference image, with a smaller value indicating that the fused image is more similar to the reference image. It is defined as follows:

$$\text{RMSE} = \sqrt{\frac{1}{HWB}\sum_{i=1}^{H}\sum_{j=1}^{W}\sum_{b=1}^{B}(X(i,j,b) - Y(i,j,b))^2} \tag{20}$$

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TABLE II
EXPERIMENTAL RESULTS OF DIFFERENT FUSION METHODS ON THE PAVIAU DATASET. (RED, GREEN, AND BLUE INDICATE THE BEST, THE SECOND-BEST, AND THE THIRD-BEST PERFORMANCE, RESPECTIVELY)

Ratio	Metrics	ICCV15	CNMF	HySure	SSRNET	HyperKite	MoGDCN	TSFN	3DT-Net	QIS-GAN
×4	RMSE	3.4703	2.4449	2.9139	**1.9977**	2.1626	2.3847	**1.8993**	2.3711	**1.8923**
	PSNR	37.008	40.050	38.526	**41.805**	41.116	40.267	**42.244**	40.317	**42.276**
	ERGAS	2.4315	1.7243	2.1350	**1.4284**	1.5465	1.6871	**1.3295**	1.6588	**1.3432**
	SAM	3.4684	2.4642	3.1410	**2.1863**	2.3391	2.4310	**2.0554**	2.5258	**2.0611**
×8	RMSE	5.6107	5.9873	6.2283	**2.0764**	2.5904	2.2609	**2.0498**	2.1537	**2.0115**
	PSNR	32.835	32.271	31.928	**41.470**	39.548	40.730	**41.582**	41.152	**41.746**
	ERGAS	3.6163	3.6373	3.8011	**1.4742**	1.7962	1.5771	**1.4568**	1.5195	**1.4401**
	SAM	4.8763	5.3929	6.4600	**2.2752**	2.7231	2.4685	**2.2372**	2.3473	**2.1691**
×16	RMSE	11.3288	18.9683	20.1928	**2.2017**	2.9709	2.7195	2.3246	**2.1518**	**2.0900**
	PSNR	26.732	22.255	21.712	**40.960**	38.358	39.126	40.489	**41.160**	**41.413**
	ERGAS	7.5495	11.9394	12.5531	**1.5591**	2.0355	1.8747	1.6569	**1.5230**	**1.4777**
	SAM	6.6987	12.0202	14.3640	**2.3946**	3.0739	2.8860	2.4863	**2.3802**	**2.2820**

where X is the fused image and Y is the reference image. H, W, and B are the height, width, and number of bands of the image, respectively.

PSNR is used to measure the ratio between the signal intensity and the noise. A higher PSNR indicates a smaller difference between the reconstructed image and the original image, which means that the quality of the reconstructed image is higher. The expression for PSNR is

$$\text{PSNR} = \frac{1}{B}\sum_{i=1}^{B} 10 \cdot \log_{10}\left(\frac{\max(Y_i)^2}{\text{mse}(X_i, Y_i)}\right) \tag{21}$$

where mse is the mean squared error.

ERGAS combines spatial and spectral information errors to assess the overall performance of the fusion algorithm, with values closer to zero indicating better fusion quality, defined as

$$\text{ERGAS} = \frac{100}{r}\sqrt{\frac{1}{B}\sum\nolimits_{i=1}^{B}\left(\frac{\text{RMSE}(X_i, Y_i)}{\overline{Y_i}}\right)^2} \tag{22}$$

where r represents the rate of downsampling.

SAM reflects the angle between the spectral vectors of the fused image and the reference image and is used to evaluate the degree of spectral distortion during the fusion process, with a smaller value indicating a better fusion result, defined as

$$\text{SAM} = \frac{1}{HW}\sum_{i,j}\arccos\left(\frac{Y_{ij}^T \cdot X_{ij}}{\|Y_{ij}\|_2 \|X_{ij}\|_2}\right). \tag{23}$$

D. Experimental Results

We compared QIS-GAN with three traditional fusion algorithms, ICCV15 [37], CNMF [36], and Hysure [67], as well as four CNN-based deep-learning algorithms, SSRNET [9], HyperKite [68], MoGDCN [69], TSFN [70], and Transformer-based deep-learning 3DT-Net [71] in fusion experiments with ×4, ×8, and ×16 times scale factors. We aimed to demonstrate the superiority of QIS-GAN.

1) Results on the PaviaU Dataset: Table II presents the quantitative evaluation results of the QIS-GAN and the comparison algorithms on the PaviaU dataset at different scale factors. It can be observed that QIS-GAN performs best in all evaluation metrics at ×8 and ×16 scale fusion while achieving the second-best ERGAS and SAM metrics at ×4 times fusion. The reason for this phenomenon is that QIS-GAN uses two QIS modules for upsampling, which weakens the mapping between images to some extent under the fusion of a ×4 times resolution difference. However, the use of three and four QIS modules at ×8 and ×16 times scale factors, respectively, enhances the fitting ability of QIS-GAN, and the advantages of QIS-GAN in fusion become more prominent as the fusion scale factor increases. Overall, QIS-GAN exhibits better reconstruction quality on this dataset. TSFN and SSRNET, respectively, achieve the second and third-best results in ×8 times resolution fusion, while 3DT-Net achieved the second-best performance at ×16 times scale factor, which may be due to the different architectures and parameter settings of the models. Fig. 5 shows the fusion results of each algorithm. Due to space limitations, we only display the image at ×8 times scale factor. It can be seen from the figure that all algorithms can significantly improve the spatial resolution of LR-HSI. However, it is difficult to distinguish the quality of fusion results visually. Therefore, we display the SAM and mean relative absolute error (MRAE) heatmaps of each fusion result at each pixel. SAM reflects the spectral performance of the reconstructed image, while MRAE displays the overall performance of the reconstructed image. From the two types of heatmaps, it can be clearly seen that in the case of an ×8 times resolution difference, the fusion quality of traditional algorithms lags far behind that of deep-learning-based algorithms. This is because there are a large number of nonlinear transformations in the fusion of LR-HSI and HR-MSI, and the powerful nonlinear modeling ability of deep learning can better fit effective fusion parameters. For deep-learning algorithms, due to the high spatial resolution of PaviaU and the small number of mixed pixels, the fusion quality of each algorithm is relatively close, but the quantitative evaluation proves that QIS-GAN outperforms the compared algorithms.

2) Results on the Botswana Dataset: The Botswana dataset has a spatial resolution of 30 m and contains 145 bands, which is more than the PaviaU dataset in terms of band and mixed pixel count. Therefore, fusion on this dataset is more challenging. Table III shows the quantitative evaluation of various algorithms on the Botswana dataset. QIS-GAN achieved the third-best performance in ×4 times fusion on this dataset, but its performance was still strong compared to the optimal result in all indicators. However, at ×8 and

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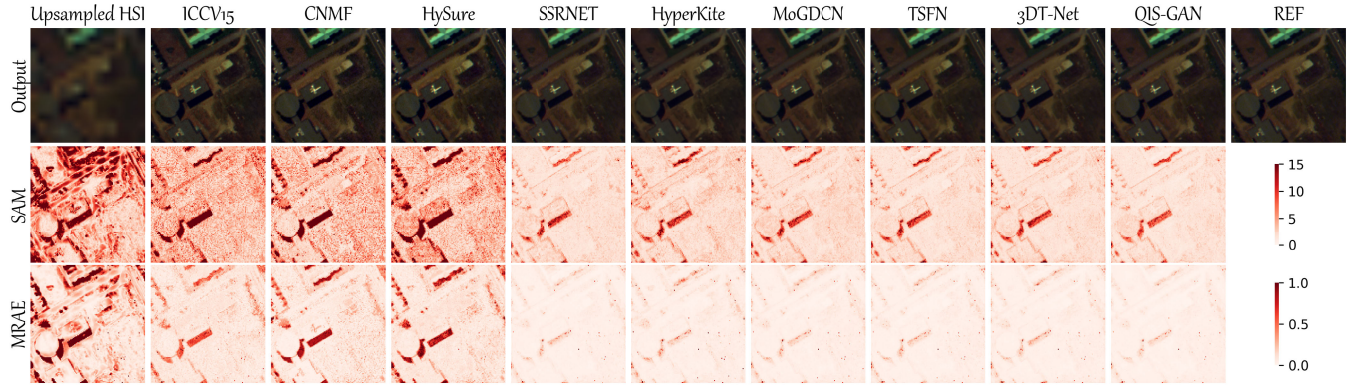


Fig. 5. Fusion results of the PaviaU dataset. (The reconstructed fusion image is presented in the first row, where "REF" represents the reference image. The R-G-B channels correspond to band 67-29-1. The second row displays the SAM error heatmap, which illustrates the quality of spectral reconstruction on a pixel-wise level. The third row exhibits the MRAE heatmap, which demonstrates the magnitude of pixel-wise errors.)

TABLE III

EXPERIMENTAL RESULTS OF DIFFERENT FUSION METHODS ON THE BOTSWANA DATASET. (RED, GREEN, AND BLUE INDICATE THE BEST, THE SECOND-BEST, AND THE THIRD-BEST PERFORMANCE, RESPECTIVELY)

Ratio	Metrics	ICCV15	CNMF	HySure	SSRNET	HyperKite	MoGDCN	TSFN	3DT-Net	QIS-GAN
×4	RMSE	0.5364	0.5929	0.5415	0.6354	**0.4852**	0.5175	**0.4848**	0.5273	**0.4876**
	PSNR	36.637	35.767	36.554	35.166	**37.508**	36.949	**37.515**	36.785	**37.465**
	ERGAS	2.0877	2.0628	2.1527	5.9588	**1.6026**	2.5780	**1.6526**	3.2361	**1.7554**
	SAM	2.2605	2.3080	2.2475	2.8087	**1.9905**	2.1405	**1.9925**	2.2391	**2.0027**
×8	RMSE	0.7576	0.7978	0.7625	0.6767	**0.5258**	0.5828	**0.5091**	0.5296	**0.5037**
	PSNR	33.638	33.188	33.582	34.619	**36.810**	35.917	**37.090**	36.748	**37.182**
	ERGAS	2.6813	2.4863	3.0383	9.8694	**1.7888**	3.5283	**1.6832**	3.2486	**1.8678**
	SAM	3.1876	3.1864	3.2312	3.0490	**2.2259**	2.5494	**2.1285**	2.3173	**2.1266**
×16	RMSE	1.3392	3.2570	4.1224	0.6724	**0.5797**	0.5967	**0.5369**	0.6268	**0.5202**
	PSNR	28.690	20.970	18.924	34.674	**35.963**	35.711	**36.628**	35.284	**36.903**
	ERGAS	4.1715	7.0004	7.8074	7.0053	**2.0410**	3.7135	**1.8721**	5.3961	**1.9082**
	SAM	4.9807	6.0142	12.3668	3.0283	**2.5243**	2.6419	**2.2472**	2.8269	**2.2157**

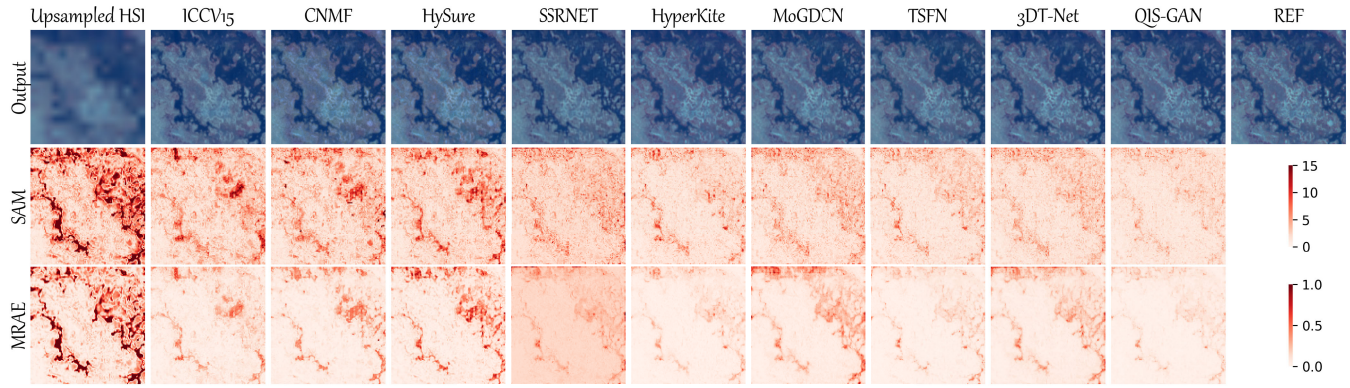


Fig. 6. Fusion results of the Botswana dataset. (The reconstructed fusion image is presented in the first row, where "REF" represents the reference image. The R-G-B channels correspond to band 48-15-4. The second row displays the SAM error heatmap, which illustrates the quality of spectral reconstruction on a pixel-wise level. The third row exhibits the MRAE heatmap, which demonstrates the magnitude of pixel-wise errors.)

×16 times scales, QIS-GAN performed best in all indicators except ERGAS. This may be because Botswana has more mixed pixels, making the nonlinearity of the fusion process higher. QIS-GAN only contains two convolutional layers to fit the nonlinear mapping of the image, resulting in slightly lower nonlinear modeling ability. Overall, QIS-GAN is still competitive for this data distribution. For all three scaling factors, TSFN and HyperKite ranked second and third as a whole, respectively. SSRNET was better than HyperKite on the PaviaU dataset, which shows that the performance of deep-learning algorithms depends not only on the model itself, but also on the data distribution. It is worth noting that SSRNET performed poorly in ERGAS, and even far below traditional algorithms at ×4 and ×8 times scales, indicating that SSRNET was trapped in local optimality, resulting in inferior fusion quality. From the heatmap in Fig. 6, it can be seen that the color of QIS-GAN is generally lighter, indicating that its fusion result is closest to the reference image.

3) Results on the Washington Dataset: This section analyzes the quantitative evaluation and fusion results of each algorithm on the Washington dataset by presenting Table IV and Fig. 7. The spatial resolution of this dataset is 2.5 m, and

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TABLE IV

EXPERIMENTAL RESULTS OF DIFFERENT FUSION METHODS ON THE WASHINGTON DATASET. (RED, GREEN, AND BLUE INDICATE THE BEST, THE SECOND-BEST, AND THE THIRD-BEST PERFORMANCE, RESPECTIVELY)

Ratio	Metrics	ICCV15	CNMF	HySure	SSRNET	HyperKite	MoGDCN	TSFN	3DT-Net	QIS-GAN
×4	RMSE	2.5517	4.8310	4.7519	2.7692	**2.4798**	3.0107	**1.9711**	3.3828	**1.9550**
	PSNR	39.075	33.531	33.675	38.365	**39.324**	37.639	**41.318**	36.627	**41.389**
	ERGAS	**1.6123**	1.7598	1.6377	1.8532	1.6784	2.0528	**1.3068**	2.2717	**1.3057**
	SAM	1.6822	2.6602	2.6550	1.8282	**1.5747**	1.9340	**1.2573**	2.1650	**1.2706**
×8	RMSE	5.0676	12.6844	12.0797	**2.7541**	2.9283	3.6653	**2.3173**	3.2261	**2.1640**
	PSNR	33.116	25.147	25.571	**38.412**	37.880	35.930	**39.912**	37.038	**40.507**
	ERGAS	2.9651	5.1693	3.8658	**1.7579**	1.9844	2.4008	**1.4957**	2.1919	**1.4241**
	SAM	3.0583	5.3424	4.7346	**1.8127**	1.8940	2.3413	**1.5090**	2.0796	**1.4143**
×16	RMSE	9.7494	19.2730	18.7587	3.0157	3.5491	3.3766	**2.6310**	**2.8914**	**2.4306**
	PSNR	27.432	21.513	21.748	37.624	36.210	36.642	**38.810**	**37.990**	**39.498**
	ERGAS	5.6769	9.8125	10.8746	**1.9530**	2.3920	2.2610	**1.7190**	2.0432	**1.6590**
	SAM	5.5360	8.9242	9.6310	2.0012	2.3050	2.2066	**1.6956**	**1.9073**	**1.5892**

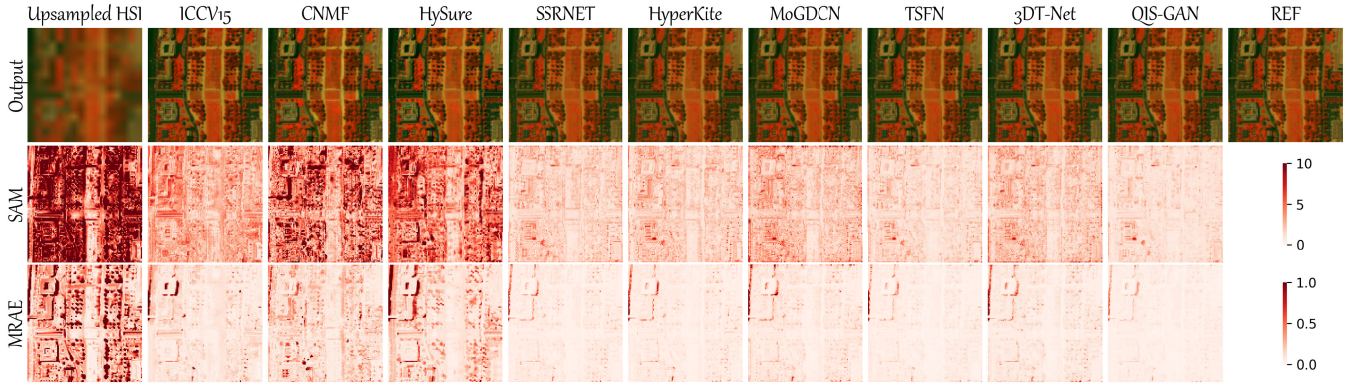


Fig. 7. Fusion results of the Washington dataset. (The reconstructed fusion image is presented in the first row, where "REF" represents the reference image. The R-G-B channels correspond to band 79-49-19. The second row displays the SAM error heatmap, which illustrates the quality of spectral reconstruction on a pixel-wise level. The third row exhibits the MRAE heatmap, which demonstrates the magnitude of pixel-wise errors.)

the degree of mixed pixels is relatively low compared to the Botswana dataset. However, the dataset contains 191 bands, making the fusion experiment challenging. From the quantitative evaluation and fusion results, QIS-GAN performed best on all three scales. Although it ranked second in SAM at ×4 times scale, it still had strong competitiveness. From the heatmap in Fig. 7, it is evident that QIS-GAN has the lightest color, indicating that it is closest to the reference image. CNMF and HySure among traditional algorithms performed poorly on this dataset, which may be because the dataset contains more bands, and CNMF and HySure cannot effectively characterize the mapping, resulting in a decrease in generalization ability. It is worth noting that the effectiveness of traditional algorithms decreases significantly as the scaling factor increases. This is because with the increase in resolution difference, the nonlinearity mapping becomes more complex, and traditional algorithms' nonlinear modeling ability is far inferior to deep-learning algorithms. SSRNET performed well when the degree of mixed pixels was low, ranking third in ×8 and ×16 times fusion. The TSFN algorithm still had the best fusion accuracy except for QIS-GAN. It can be concluded that even with a large number of bands, QIS-GAN can still fit more effective mapping to achieve the fusion of LR-HSI and HR-MSI.

4) Results on the Houston Dataset: The Houston dataset has the same spatial resolution as the Washington dataset, but has fewer bands, making fusion slightly easier than the Washington dataset. Table V and Fig. 8 show the quantitative evaluation and fusion results of algorithms on the Houston dataset. The results indicate that QIS-GAN had the best fusion effect at all three scaling factors, while the CNMF algorithm also achieved the second-best performance at ×4 times fusion. This may be due to the data distribution of the Houston dataset, but as the scaling factor increases, the effectiveness of its algorithm decreases significantly. From the fusion result images, it can be seen that the heatmap of QIS-GAN has the lightest color overall, and it is closer to the reference image both in terms of spectral and overall image reconstruction, once again confirming that QIS-GAN is superior to other algorithms.

E. Comparative Analysis of Spectral Curves

In this section, we plot the spectral curves of a randomly selected pixel on each dataset (see Fig. 9), and it can be observed that the fusion images generated by QIS-GAN exhibit good spectral consistency. Although some deviations are noticeable in certain bands, overall, QIS-GAN performs well in preserving the spectral consistency of the generated images. This demonstrates that QIS-GAN is capable of effectively handling spectral information from different bands, resulting in higher-quality fusion images. This is of significant practical importance for the application of remote-sensing images, as it can aid in better understanding and utilization of remote-sensing images and enhance the effectiveness of remote-sensing technology in various fields.

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TABLE V

EXPERIMENTAL RESULTS OF DIFFERENT FUSION METHODS ON THE HOUSTON DATASET. (RED, GREEN, AND BLUE INDICATE THE BEST, THE SECOND-BEST, AND THE THIRD-BEST PERFORMANCE, RESPECTIVELY)

Ratio	Metrics	ICCV15	CNMF	HySure	SSRNET	HyperKite	MoGDCN	TSFN	3DT-Net	QIS-GAN
×4	RMSE	2.0067	**0.8818**	**1.0724**	1.3104	1.4339	1.4160	1.1329	1.8807	**0.8764**
	PSNR	41.689	**48.831**	**47.132**	45.391	44.608	44.717	46.655	42.252	**48.885**
	ERGAS	2.1230	**1.1701**	1.6541	4.0481	1.5814	1.8584	**1.3691**	2.4729	**1.1887**
	SAM	1.7698	**1.5224**	1.7432	2.3940	1.9566	2.2646	**1.7377**	2.8538	**1.4823**
×8	RMSE	3.5214	3.4887	3.5400	**1.3654**	1.7205	1.5567	**1.2439**	2.7103	**1.0439**
	PSNR	36.804	36.885	36.759	**45.033**	43.026	43.895	**45.843**	39.078	**47.365**
	ERGAS	3.5771	3.5561	3.8970	3.4883	**1.8653**	2.0051	**1.5155**	3.3426	**1.3832**
	SAM	4.7085	4.1988	4.6842	2.4935	**2.4117**	2.5861	**1.9200**	3.9620	**1.8229**
×16	RMSE	6.5153	8.8496	12.7253	**1.4660**	2.1919	2.0906	**1.4349**	1.8284	**1.2551**
	PSNR	31.460	28.800	25.645	**44.416**	40.922	41.333	**44.602**	42.497	**45.765**
	ERGAS	6.3952	8.6246	12.5735	**1.9011**	2.3394	2.6527	**1.7683**	2.4805	**1.5997**
	SAM	7.7006	10.2171	17.6118	**2.5660**	3.3747	3.5737	**2.3917**	3.0840	**2.2513**

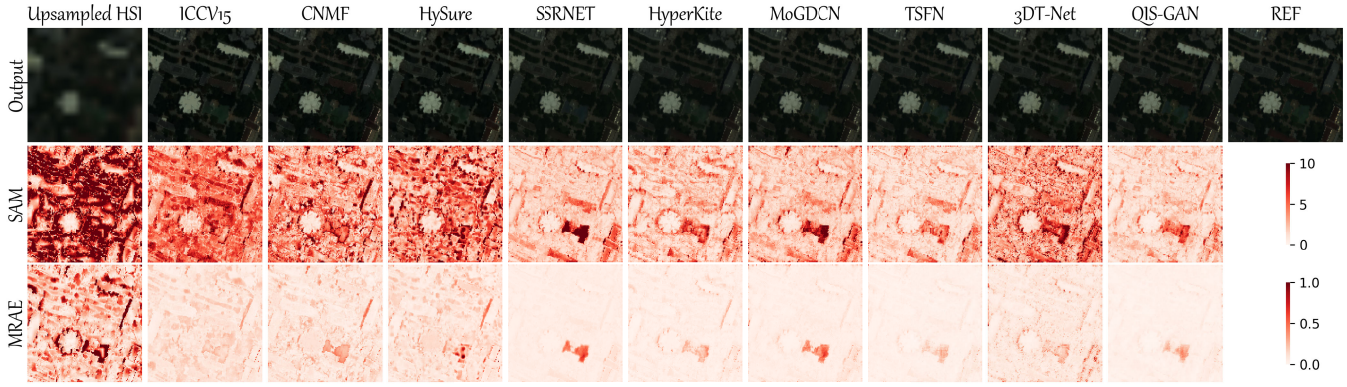


Fig. 8. Fusion results of the Houston dataset. (The reconstructed fusion image is presented in the first row, where "REF" represents the reference image. The R-G-B channels correspond to band 61-41-21. The second row displays the SAM error heatmap, which illustrates the quality of spectral reconstruction on a pixel-wise level. The third row exhibits the MRAE heatmap, which demonstrates the magnitude of pixel-wise errors.)

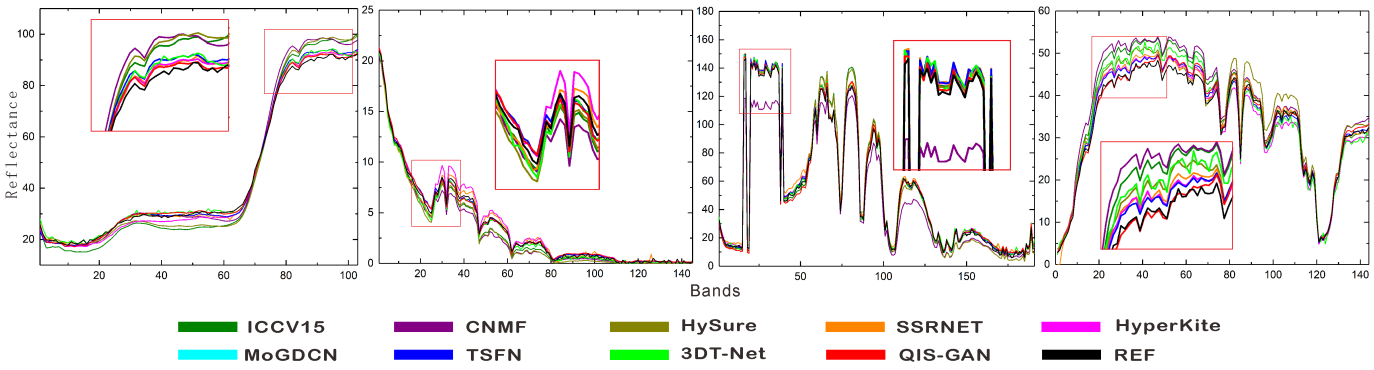


Fig. 9. Comparison of spectral reflectance on the four datasets.

F. Comparison of Different Sampling Methods

In this section, we compared the QIS with several other sampling methods, such as Bicubic, Bilinear, Transpose, Pixel Shuffle, and GINS, on the Botswana dataset. The quantitative evaluation results (see Table VI) show that the QIS performs the best in both spectral and spatial domains. This is because the method can adaptively learn the distribution of sample data and generate high-quality images based on the characteristics of the sample data, while other interpolation methods may not effectively fit the actual distribution features of the sample data, leading to poor interpolation results. In addition, the QIS utilizes the powerful expression ability of deep-learning models, which improves the interpolation performance compared to the GINS method that directly generates the target resolution. Therefore, QIS is an efficient, accurate, and highly applicable image interpolation method.

G. Ablation Experiment

In this section, we perform a series of ablation experiments on the Washington dataset with a scale factor of 8 to verify the effectiveness of QIS-GAN. Given the simplicity of the QIS-GAN structure, we focus on the ablation experiments on the discriminator, GINS, and QIS modules. Specifically, we take the following steps: 1) first, the discriminator of QIS-GAN is removed to determine its effect; 2) the GINS of QIS-GAN is deleted, and instead, LR-HSIs are directly sampled to the target resolution using implicit sampling to validate the role of the GINS; and 3) the QIS module is

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TABLE VI
EXPERIMENTAL RESULTS OF DIFFERENT INTERPOLATION METHODS. (RED INDICATES THE BEST PERFORMANCE)

Interpolation methods	RMSE	PSNR	ERGAS	SAM
Bicubic	0.6315	35.220	2.1822	2.6369
Bilinear	0.6492	34.979	2.2081	2.7013
Transpose	0.6290	35.254	2.1412	2.6757
Pixel shuffle	0.6626	34.802	2.1275	2.9270
GINS	0.5351	36.657	2.0472	2.2742
QIS	**0.5050**	**37.161**	**1.8102**	**2.1090**

eliminated and replaced it with the sampling method of JIIF to directly sample to the target resolution. The experimental results are shown in Table VII.

It can be observed that the accuracy decreases after removing the discriminator, which indicates that the introduction of the discriminator can effectively enhance the image generation capability of QIS-GAN. Whereas the deletion of GINS implies the abandonment of the guiding role of HR-MSI, the network actually realizes the single super-resolution of HR-HSI. Due to the relatively simple structure of the QIS-GAN network, the network fits a nonlinear mapping, which results in the network not being able to converge effectively, and thus the accuracy is poor. After eliminating the QIS module, the accuracy of fusion, although performing better, still has a significant gap compared with QIS-GAN, which indicates that the QIS module can effectively enhance the feature extraction capability on the basis of JIIF to achieve higher fusion accuracy. In summary, it can be concluded that all the modules of QIS-GAN bring significant gains to the MHIF task and do not produce redundancy.

H. Analysis of Model Complexity

In this section, we conducted a model complexity comparison analysis of QIS-GAN and other deep-learning algorithms on the Botswana dataset (see Table VIII) during the inference phase. From the indicators of ModelSize, Flops, and Params, it can be seen that QIS-GAN has the lowest complexity and the most lightweight effect compared to other algorithms. This is because our algorithm uses fewer convolutional kernels, and the QIS is composed of MLP, which occupies less memory. In combination with the experimental results in Section IV-D, QIS-GAN not only has lightweight features but also performs remarkably well in terms of performance. This means that our algorithm can achieve high-quality image fusion tasks on devices with lower computing resources.

I. Comparison of Data Under Different Sampling

To more comprehensively evaluate the performance of QIS-GAN and other deep-learning algorithms under different spatial degradation conditions, we generated LR-HSI using other interpolated degradation methods and performed experimental comparisons. The chosen dataset is Washington and the scale factor is set to 8. In Table IX, we show the fusion results of these methods and compare the performance of QIS-GAN with other algorithms. It can be observed that QIS-GAN shows better performance overall. Although QIS-GAN has the second-best results under bicubic interpolation, we can clearly see that the fusion accuracy of QIS-GAN is less affected by the data generated by different degradation methods. Meanwhile, TSFN also achieves better results and can overcome the effects of different degradation methods, which further validates that this two-stream architecture of TSFN is more effective in capturing the nonlinear mapping relative to the other comparison algorithms. The SSR-NET and HyperKite also show a significant ability to generalize to different degradation methods. This indicates that these algorithms can maintain stable performance under different degradation conditions. However, the performance of MoGDCN is more affected by the degradation methods, which suggests that the model is less adaptable in dealing with different degradation methods. Finally, 3DT-Net ranks last in terms of performance, probably due to its inability to effectively fit the nonlinear mapping in the Washington dataset with more bands. In summary, QIS-GAN can effectively overcome the generated data under different spatial degradation conditions and shows better generalization performance.

J. Robustness Experiments

To verify the interference resistance of QIS-GAN, we conduct robustness experiments on deep-learning algorithms under 8× scale factor on the Washington dataset. First, in the noise resistance experiment, we add Gaussian white noise with different signal-to-noise ratios (SNRs) to LR-HSI and HR-MSI, respectively, to test the resistance to noise of algorithms. The experimental results (see Table X) show that QIS-GAN still exhibits the best image fusion effect under the three SNR noise pollution. Second, we conduct an antispectral mutation experiment to simulate the spectral mutation by randomly disrupting the band alignment of the images to be fused. The experimental results show that QIS-GAN still exhibits good robustness even in the face of the occurrence of spectral information interference. With the results of the antinoise experiment and the antispectral mutation experiment, we can conclude that QIS-GAN has excellent robustness and is suitable for image fusion tasks in various complex environments.

K. Sensitivity Analysis for Different Image Sizes

In this section, we count the flops of QIS-GAN and other deep-learning algorithms for different fused image sizes. The image with 145 bands and the 8× spatial resolution scale was selected for testing. With the statistics in Table XI, we can clearly see that QIS-GAN performs optimally in terms of flops of all sizes. It is worth noting that the increase in flops of QIS-GAN is not significant as the input size changes at the inference phase. This means that our algorithm does not introduce too much computational burden when dealing with inputs of larger sizes. In contrast, other algorithms may face greater computational complexity when dealing with inputs of larger sizes. These results further demonstrate that QIS-GAN has a significant degree of lightweight at different sizes. QIS-GAN can keep a relatively stable number of flops, both

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TABLE VII

ABLATION STUDY RESULTS OF THE PROPOSED ELABORATE DESIGNS IN THE PROPOSED QIS-GAN. (RED INDICATES THE BEST PERFORMANCE)

Methods	Discriminator	GINS	QIS	RMSE	PSNR	ERGAS	SAM
N_D	✗	✓	✓	2.3032	39.966	1.5282	1.5083
N_G	✓	✗	✓	18.6837	21.783	9.7171	9.5668
N_Q	✓	✓	✗	2.3871	39.655	1.6306	1.5377
QIS-GAN	✓	✓	✓	**2.1640**	**40.507**	**1.4241**	**1.4143**

TABLE VIII

MODEL SIZE, FLOPS, AND PARAMS OF THE QIS-GAN AND THE OTHER DEEP-LEARNING MODELS DURING THE INFERENCE PHASE. (RED INDICATES THE BEST PERFORMANCE)

Methods	ModelSize(M)	Flops(G)	Params(M)
SSRNET	2.17	9.3008	0.57
HyperKite	2.21	219.4157	0.57
MoGDCN	53.67	470.4815	14.06
TSFN	8.15	34.4237	2.10
3DT-Net	40.50	411.7508	10.50
QIS-GAN	**1.46**	**7.7500**	**0.38**

TABLE IX

PSNR COMPARISON OF DATA UNDER DIFFERENT SAMPLING. (RED INDICATES THE BEST PERFORMANCE)

Degradation methods	LANCZOS4	BICUBIC	NEAREST
SSRNET	37.831	37.855	37.682
HyperKite	38.785	38.808	38.600
MoGDCN	30.238	33.108	31.386
TSFN	40.225	**40.151**	39.912
3DT-Net	32.569	33.073	32.278
QIS-GAN	**40.325**	40.137	**40.406**

TABLE X

PSNR STATISTICS FOR ROBUSTNESS EXPERIMENTS ON THE WASHINGTON DATASET. (RED INDICATES THE BEST PERFORMANCE)

Methods	SNR 10%	SNR 20%	SNR 30%	Spectral Mutation
SSRNET	30.827	39.636	40.185	40.084
HyperKite	30.856	39.412	39.979	39.997
MoGDCN	31.039	39.518	40.046	40.007
TSFN	31.057	39.738	39.829	40.056
3DT-Net	31.122	39.558	39.884	40.185
QIS-GAN	**31.346**	**39.836**	**40.235**	**40.226**

TABLE XI

FLOPS (G) STATISTICS FOR DIFFERENT FUSED IMAGE SIZES. (RED INDICATES THE BEST PERFORMANCE)

Size	32	48	64	96
SSRNET	0.5813	1.3079	2.3252	5.2317
HyperKite	13.7135	30.8553	54.8539	123.4213
MoGDCN	29.4051	66.1615	117.6204	264.6458
TSFN	2.1515	4.8408	8.6059	19.3633
3DT-Net	25.7344	57.9025	102.9377	231.6098
QIS-GAN	**0.4843**	**1.0897**	**1.9372**	**4.3587**

when dealing with fused images of smaller sizes and when dealing with fused images of larger sizes.

V. CONCLUSION

This study proposes a lightweight LR-HSI and HR-MSI fusion network called QIS-GAN and innovatively proposes a QIS based on the INR joint Quadtrees idea, which recovers more accurate images by sampling the pixel values of more leaf nodes. We conducted experiments with four, eight, and 16 times the difference in resolution, respectively, and extensive experiments show that QIS-GAN achieves a high-precision fusion of LR-HSI and HR-MSI.

Unfortunately, our article did not delve into exploring the impact of coordinate grids on the network, which is also worth investigating and exploring. Additionally, global features have shown promising results in image restoration tasks, so further investigating nonlocal methods for handling MHIF tasks based on INR can be beneficial. On the other hand, it is possible to design more data-driven approaches tailored to MHIF tasks, leveraging the data itself to design INR methods. Furthermore, we will continue along the current line of thinking to address the shortcomings of INR and apply it to the field of remote sensing.

DECLARATION OF COMPETING INTEREST

No potential conflict of interest was reported by the authors.

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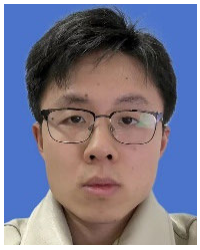
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